

Chapter 127 — Male-to-Female Gender-Affirming Surgery: Vaginoplasty

INTRODUCTION

- Increased social acceptance has produced **increased access to care for transgender patients**. Urologists **align physical anatomy with gender identity while maintaining physiologic function** — the dual objective of the whole chapter.

PREPARATION FOR SURGERY

Mental Health Evaluation

- Mental health services serve **two distinct functions**: management of dysphoria itself, and **assessment of surgical readiness**. The second has direct operative consequences.
- **Early involvement is justified by the long lead time** — preparation for vaginoplasty takes months, giving time to identify and address **psychosocial barriers of employment, housing and social support**.
- Interventions before surgery: **smoking cessation, weight loss, and brief solution-focused supportive work** to establish psychosocial stability.
- Mental health professionals also **equip patients with healthy coping skills for the psychological distress and pain of recovery**, particularly when complications occur.
- Because of discrimination and other stressors, **psychiatric diagnoses are highly comorbid with gender dysphoria**. **Comorbid conditions should be treated before gender-affirming surgery wherever possible**, as they adversely affect postoperative recovery.
- **Decisions to delay surgery until psychiatric or psychosocial stability improves are case-by-case and made in collaboration with the patient**, never categorical.

Risk Framework

- Risk related to psychosocial and psychiatric stability resolves into **two categories**: (1) **non-adherence to postoperative treatment** — in practice, failure of dilation; and (2) **worsening psychosocial or psychiatric prognosis after surgery**.
- **The nature and degree of risk are not static** — a range of individual risk and protective factors in the patient's biopsychosocial context must be weighed.
- **The critical logical point**: where **depression and suicidal symptoms are largely driven by dysphoria**, treatment of the dysphoria becomes **more critical**, and it is reasonable to expect surgery to **reduce distress and improve function**. Psychiatric symptoms are not in themselves a reason to defer.
- Factors to assess: **history of adherence concerns; safety in environment and relationships; help-seeking; health literacy; access to financial resources; social and instrumental support; adaptive or maladaptive coping including substance use; rapport and trust with the medical team and institution; impulsivity; distress tolerance; resource use; system navigation competencies; caregiving needs; self-advocacy; and involvement of support persons**.

Updated World Professional Association for Transgender Health Criteria

- Earlier **World Professional Association for Transgender Health (WPATH)** guidance was criticised for **excessive gatekeeping creating unnecessary barriers** to gender-affirming care including vaginoplasty.

Standards of Care version 8 (SOC8) emphasises access to care and an informed consent approach. Six specific changes:

- **One assessment rather than two**, justified by **high rates of agreement among independent assessors**.
- **Who may assess has widened**. No longer confined to licensed mental health professionals: the assessor must hold a **licence and at minimum a master's degree**, be **competent in transgender health**, undertake **continuing education**, and have **mental health evaluation within their scope of practice** (Statement of Recommendation 5.1). **Primary care providers and surgeons may be appropriate assessors** depending on experience and expertise. Mental health professionals become **consultants to an interdisciplinary team rather than the sole determinants of medical necessity**.
- **Six months of hormone replacement therapy (HRT) before gonadectomy**, consistent with the patient's goals, is **sufficient** — the rationale being **experience of the effect of testosterone blockers before irreversible changes in hormone production**.
- **Psychiatric and medical conditions need no longer be well managed before medical transition**; instead **risks and benefits are identified and discussed** — the shift from prerequisite to disclosure.
- **The 12-month social transition or "experience in a role" requirement is removed**, having been a **serious barrier for people facing significant threats of violence for expressing their gender**. Discussion of social transition is the patient-centred alternative; transition is nuanced and not dictated by providers.
- **Capacity to consent is procedure-specific**, with emphasis where **fertility is affected**. Rarely, **intellectual or developmental disability, severe psychiatric symptoms such as psychosis, or cognitive impairment** may **temporarily or permanently** affect an adult's capacity. Where capacity is in doubt, psychologists help construct a plan to **support or restore capacity**, which may include **formal or informal proxy decision makers** in the consent process.

Concept — *SOC8 substitutes information the patient must receive for thresholds the patient must clear: two assessors become one, managed comorbidity becomes disclosed comorbidity, and mandated real-life experience becomes discussed social transition. Capacity and fertility counselling remain hard obligations because they alone are irreversible.*

Care Coordination for Gender-Affirming Surgery

- **Postoperative depression is not uncommon and may severely affect recovery and vaginal dilation** — the direct link from mood to surgical outcome.
- Patients must **identify sources of social support and a plan for postoperative care** before surgery, and **ready access to medical care in an emergency** must be confirmed.
- **For the first 3 weeks after surgery the patient should be within one hour's driving distance of the operating hospital**.
- **Misgendering may occur throughout the health care system and is a source of dissatisfaction**. The electronic health record must reflect the patient's gender and preferred name.
- **Financial administrative staff require education** on the surgery and WPATH guidelines to obtain **prior authorisation** and bill appropriately.
- **The surgeon should be deeply involved in the initial stages** — confirming the WPATH criteria are being met and that insurance preauthorisation is correctly submitted.

Weight Limitations

- **Body mass index (BMI) is the major factor affecting wound healing and tension on the closure**.
- **Superficial wound healing problems are cited across a wide range, from 3.3% to 33%**.
- **There is no well-established BMI cut-off; in practice many providers require a BMI of 35 kg/m² or less**.

- **A higher BMI may be appropriate** based on provider comfort **and if the patient does not desire a vaginal canal** — that is, for vulvoplasty.
- **Very low BMI may signal an eating disorder**, itself affecting wound healing.

Hair Removal

- **Scrotal skin is commonly used to line the neovaginal canal**, so **permanent hair removal by laser or electrolysis prevents hair growing within the canal after vaginoplasty**. Hair removal is performed **before** surgery.
- **Intraoperative follicle removal alone may be insufficient**, because **hair growth occurs in several phases** — the reason preoperative treatment cannot be substituted for at the table.
- **Intravaginal hair growth is a common long-term complaint** and leads to **infection, hairballs, and trapping of epidermal debris and lubricant**.
- **Hair removal template (Fig. 127.1):** the **scrotal and perineal skin** that will form the full-thickness skin graft lining the canal.

Perioperative Hormone Management

- **Venous thromboembolism (VTE) risk may be increased on hormone therapy**, especially with **malignancy, obesity and tobacco use**.
- **Bottom line of the 2022 meta-analysis:** continuing oestrogen therapy perioperatively incurs no additional VTE risk in vaginoplasty.
- **Discontinuation carries its own harm** — mental distress.
- **Practice:** continue hormone therapy through the perioperative period.

Orchiectomy

- **Gender-affirming orchiectomy may be the sole genital surgery or the first stage**, for several reasons: some patients' **dysphoria relates primarily to seeing and feeling the testicles**, and removal satisfies them; it **facilitates tucking of the phallus for concealment**; and it **minimises the hormone regimen**.
- **Bottom line of the hormonal effect:** after orchiectomy, **essentially all patients were able to stop spironolactone**, most **discontinued progesterone**, and a substantial minority **reduced their oestradiol dose**.
- **All patients undergo fertility counselling before orchiectomy**, with referral for sperm banking before surgery if preservation is wanted. **It must be made clear that orchiectomy and vaginoplasty lead to infertility**, since misconceptions are common.
- **Donor-site controversy:** prior orchiectomy may increase the chance of needing extragenital skin grafts at later vaginoplasty. **Peritoneal flaps and allografts reduce that need**.
- **Incision choice is the examinable point.** Inguinal incisions may compromise the blood supply of the **penile flap used to create the vulva** at eventual vaginoplasty, whereas a **scrotal incision is incorporated into the skin graft used for vaginoplasty**.
- **Therefore:** bilateral orchiectomy through a vertical midline scrotal incision.

Concept — *Orchiectomy is never standalone in planning terms: it is either the endpoint or stage one of a vaginoplasty, and is performed as though vaginoplasty will follow. That single assumption dictates the midline scrotal incision and the preservation of peri-cord tissue for later labial bulk.*

VAGINOPLASTY

Positioning and Set-Up

- **Dorsal lithotomy**, with careful padding of the **lateral aspects of the calf and the posterior heel** to minimise neuropraxia; **specific vigilance for common peroneal nerve injury** in lithotomy.
- For the robotic component, **port setup as for prostatectomy**, in **steep Trendelenburg**.
- **VTE prophylaxis: weight-dosed subcutaneous heparin plus a sequential compression device.**
- The procedure should not incur major blood loss, but **bleeding is front-loaded** — concentrated in **penile disassembly, urethrectomy/urethroplasty and vaginal canal dissection**. For this reason **tranexamic acid 1 g intravenously** is given at the beginning of the procedure.

Perineal Approach

- **Penile inversion vaginoplasty (PIV) via the perineal approach is the commonest method of gender-affirming genital reconstruction for transgender women** and supplies most of the published data, even where authors now favour robotic abdominal dissection with peritoneal flaps.
- **Goal: a functional and aesthetic organ that alleviates gender dysphoria.**
- **Homologous tissues (Fig. 127.3)** are the conceptual spine of the operation and are set out in the takeaway to this section.

Marking and Scrotal Skin Harvest

1. **Place a Foley catheter and drain the bladder**, then **plug the catheter** so the penis can be moved during disassembly.
2. **Measure the genital anatomy before incising. A short penile shaft results in shorter labia minora but does not affect canal depth** — an important counselling point.
3. **Mark the tendons of the adductor longus bilaterally** as the landmark for **clitoral position**, and **mark the midline on the dorsal penile skin to avoid torsion of the skin.**
4. **Mark and harvest the scrotal skin for the full-thickness graft that lines the neovaginal canal.** A small **rhomboid posterior flap** is raised at this incision, later **advanced into the introitus to take tension off the posterior closure.**
5. **Defat and thin the graft on the back table**; perform **hair follicle removal and fulguration**; **preserve the subcutaneous tissue beneath the scrotal skin** during harvest.
6. **Tubularise the thinned full-thickness graft over a vaginal dilator**, then use it to **augment the distal portion of the penile skin** that will be inverted into the canal.

Orchiectomy at the Time of Vaginoplasty

- Performed now if not previously done. **Retract the testicles with a towel clamp and dissect the spermatic cords proximally to the level of the inguinal canal.**
- **Leave the tissue and fat surrounding the spermatic cord behind** — it provides volume to the labia majora.

Penile Disassembly

1. **Circumcising incision**, sparing an adequate mucosal collar that will serve as the **clitoral hood**.
2. **Elevate the proximal penile skin off Buck's fascia while keeping the dartos with the penile skin.** Preserving the dartos maintains blood supply to the skin flap once it is rearranged to form the labia.
3. Dissection **between the dartos and Buck's fascia** yields a **cylindrical penile skin flap**.
4. **Extend the dissection laterally and mobilise the surrounding tissue** to produce the flap used for the labiaplasty.

Corporal Resection and Clitoroplasty

- The clitoris is created from the glans penis attached to the neurovascular bundle.
- Two techniques of separating corporal tissue from the neurovascular bundle — the examinable pair:
 - **Separate the bundle from the tunica albuginea of the corpora cavernosa:** incise the dartos layer and Buck's fascia longitudinally, with the **initial incision along the lateral side of the urethra to minimise violation of the bundle**, dissecting along the tunica until the entire bundle is elevated.
 - **Maintain the bundle in continuity with the dorsal tunica albuginea — the authors' preference**, because it **minimises handling of the neurovascular bundle and decreases the risk of nerve damage**.
- Technique for the preferred method: **open the corpus cavernosum longitudinally from glans to base ventrally → excavate the spongy tissue with a Molt periosteal elevator → coagulate the cavernosal arteries → close the proximal tunica from proximal to distal up to the level of the adductor longus tendon.**
- **Complete excision of erectile tissue is mandatory:** residual erectile tissue causes **engorgement during arousal and obstruction of the vaginal canal.**
- **Clitoroplasty:** mark the proximal glans with a "W" → **excise the ventromedial glans tissue, preserving the lateral wings → de-epithelialise the wings and suture them together to create the clitoris.**
- **The size trade-off:** preserve enough glans for an **aesthetic, appropriately sized clitoris with the maximal number of nerve endings**, but **too much preserved glans yields a prominent clitoris resembling a glans penis.**
- **The mucosal collar of distal penile skin becomes the clitoral hood.**
- After creation of the clitoris, the **neurovascular bundle is accorded onto itself and secured to the suprapubic abdominal fascia, laid flat**, with the clitoris positioned **at the level of the adductor longus tendons.**
- **The mucosal collar of preputial skin is then secured inferiorly along the urethral plate to form the medial aspect of the labia minora.**

Urethroplasty

1. **Mobilise the urethra from the distal pendulous portion proximally.**
2. **Discard the entire pendulous urethra and the distal bulbar urethra above the level of the adductor longus tendon.**
3. **Elevate the bulbospongiosus muscle — the dissection resembles urethroplasty, tracking the muscle ventrally then laterally and posteriorly, but here the muscle is excised rather than retracted. Debulking reduces obstructing tissue at the vaginal introitus.**
4. **Keep the bulbospongiosus intact until canal dissection is complete — it provides an additional layer of coverage in the rare instance of rectal injury.** This is the reason for the sequencing.
5. **Spatulate the urethra ventrally and downward to allow a straight urinary stream in the sitting position.**
6. **A 3 cm dorsal segment of bulbar urethra creates the mucosal vestibule of the vagina**, secured as a flat plate to the **inferior segment of the neoclitoris** and laterally to the **medial labia minora. Oversee the lateral edges of the urethra** to reduce significant bleeding.
7. **Mature the urethra to the inverted penile tissue** after confirming the opening is patent and the stream trajectory correct, assessed by **passing a Foley catheter or a bougie dilator.**

Neovaginal Canal Creation and the Vulvoplasty Decision

- **Thorough preoperative discussion of patient preferences is critical, because of the commitment to lifelong dilation.**

- **Vulvoplasty** — also termed **shallow, limited or zero-depth vaginoplasty** — creates the vulva but not the vaginal canal, and suits patients certain they do not wish to engage in receptive intercourse.
- **The decision is not freely reversible:** preferences may change after surgery; **vulvoplasty discards the scrotal skin that would have lined the canal;** and **later full-depth vaginoplasty is complicated by scarring of the perineal dissection and the need for additional graft tissue.**
- **Vulvoplasty reduces risk** of rectal or bladder injury and **avoids the need for abdominal access** for robotic or laparoscopic peritoneal flap harvest — relevant with **multiple comorbidities.**
- **Prior radical prostatectomy is an indication for vulvoplasty,** since **canal dissection may injure the external urinary sphincter** and cause incontinence.

Canal Dissection

1. **Initiate along the urethra by division of the central tendon,** then carry the dissection **proximally toward the prostate.** Canal dissection requires in-depth familiarity with perineal and pelvic anatomy.
2. **Avoid injury to the rectum and urethra. Drape so that digital rectal examination can be performed routinely** to evaluate the thickness of tissue between the neovaginal canal and the rectum; **a latex rectal probe cover inserted into the rectum and secured to the skin or drape** allows easy repeated assessment.
3. **A Lowsley retractor may be used to manipulate the prostate** and aid the dissection.
4. **Sharply dissect the bulbospongiosus muscle off the urethra.**
5. **For combined abdominoperineal dissection, carry the dissection proximally to the base of the prostate** for adequate depth; the proximal canal is then **developed robotically** for further depth.
6. **For a purely perineal approach, carry the dissection more proximally along the posterior aspect of the prostate; lighted retractors help.**
7. **Preserve Denonvilliers fascia posteriorly. Violation exposes the perirectal fat, and the fascia should be closed where possible,** especially if rectal examination shows thin tissue overlying the rectum.
8. **Control bleeding with judicious bipolar cautery or suture ligation; LigaSure may also help.**
9. **Calibrate the space** to confirm adequate depth and width.
 - **If the prostatic capsule is violated** — a recognised hazard as exposure becomes difficult — **check for urethral injury by palpating for the catheter or by intraurethral irrigation,** and anticipate **significant bleeding from the prostatic stroma.**
 - **In vulvoplasty, the dissection is carried only to the membranous urethra** to give the appearance of a vaginal canal.
 - **Calibration reference: Soul Source GRS vaginal dilator #4 to the last dot — 22.9 cm long, 3.8 cm in diameter.** Previous literature places vaginal depth at **10 to 15 cm,** a figure now being revised upward with the abdominoperineal and robotic approaches.

Canal Lining

1. **Defat the scrotal skin and excise or fulgurate any remaining hair follicles** to reduce regrowth.
2. **Tubularise the scrotal skin and suture it to the distal aspect of the penile skin,** most easily **over a dilator with a running absorbable suture.**
3. **Make a midline incision on the proximal inverted penile skin** to allow **delivery of the clitoris** and an **uninterrupted path for urine flow from the urethra.**
4. **Secure the penile skin to the inferior aspect of the urethra** to prevent it recoiling upward.
5. **Pass the remaining inverted penile skin and tubularised scrotal skin into the canal. Graft contact with the surrounding canal tissue is crucial for graft take;** in perineal vaginoplasty, **vaginal packing is relied on to secure that contact.**

6. Secure the medial aspect of the labia majora to the lateral aspect of the labia minora, and carefully approximate the lateral labia majora to the defect left by the scrotal excision. Excise rugated scrotal tissue for better aesthetics.
7. Secure the fat residual after the orchidectomy to give adequate bulk to the labia majora before skin closure.
8. Close in multiple layers, with a layer of Scarpa's fascial closure before the skin; the labial edges are approximated with running sutures or interrupted mattress sutures.
9. Keep the incision at least 1 cm away from the groin crease to reduce the chance of wound dehiscence.
10. Pack the canal with antibiotic-soaked vaginal packing and apply a negative pressure dressing.

Labiaplasty

- Shape and volume depend on natal anatomy — the length of shaft skin and the prepubic adipose tissue affect labial appearance.
- Labia minora, medial aspect: the distal penile skin mucosal collar, secured to the lateral aspect of the urethral plate that forms the vestibule. A running suture achieves haemostasis on the lateral side of the urethral plate, which otherwise bleeds significantly postoperatively.
- The labia minora is accentuated with horizontal mattress quilting sutures to create a more pronounced ridge.
- The labia majora is created from the lateral scrotal skin.

Concept — Penile inversion vaginoplasty is best held as a homology map rather than a list of steps: glans → clitoris; preputial mucosal collar → clitoral hood and medial labia minora; dorsal bulbar urethral plate → vestibule; penile shaft skin → canal lining; scrotal skin → full-thickness graft lining the canal; lateral scrotal skin and peri-cord fat → labia majora. Every technical rule — preserve the dartos, spare the collar, leave the fat, excise all erectile tissue — protects the blood supply, sensation or bulk of one transferred structure.

Abdominoperineal Approach

Peritoneal Vaginoplasty

- Increasing availability of robotic surgery has made the abdominoperineal approach feasible. Often called the **Davydov peritoneal vaginoplasty**, the technique was originally described for **vaginal agenesis in cisgender women** and has since been adapted.
- The external part of the procedure is identical to perineal vaginoplasty; only the proximal canal is created by abdominal dissection.
- Rationale: historical rectal injury rates of approximately 7% during canal dissection, many attributable to poor exposure and visualisation. Creating the proximal canal robotically, in a dissection similar to radical prostatectomy, reduces the risk of rectal injury.
- Critical differences from robotic prostatectomy:
 - Start by identifying the bilateral vas deferens and tracking them medially to identify the most proximal aspect of the prostate.
 - Be vigilant for organs that mimic the landmarks — the rectum can appear to mimic the seminal vesicles.
 - After elevating the seminal vesicles, carry the dissection distally along the posterior aspect of the prostate.
 - Minimise electrocautery to decrease the risk of rectal injury.
 - Sharply incise Denonvilliers fascia and enter the plane between its two leaflets. If the correct plane is established it is bloodless and develops distally with blunt dissection.

- **Connect the robotically dissected canal with the perineally dissected canal. Good communication is critical, as the two surgeons are often in slightly different planes, which is easily addressed if identified. Calibrate for adequate depth and width.**
- **Peritoneal flap techniques — two named alternatives:**
 - **Two-flap technique:** flaps raised from the **posterior bladder wall (10 cm × 10 cm)** and **lateral to the sigmoid (10 cm × 10 cm)** to augment the **most proximal aspect of the canal**. The tubularised penile and scrotal skin is passed up by the perineal surgeon and **secured circumferentially to the peritoneum** before the proximal-most canal is closed.
 - **Peritoneal pull-through:** a **single large flap (11 cm across × 20 cm longitudinally)** along the **anterior abdominal wall, based on the bilateral deep inferior epigastric arteries**, elevated and passed through the canal to the perineal surgeon, then **secured to the introitus**.
- **The proximal peritoneal defects must be closed by the robotic surgeon to minimise the risk of hernia formation.**

Intestinal Vaginoplasty

- **Less commonly performed than penile inversion vaginoplasty.** Principal indications: **severe penoscrotal hypoplasia with insufficient scrotal tissue to line the canal**, and **canal stenosis in revision vaginoplasty**.
- **Contraindications:** **inflammatory bowel disease, extensive abdominal adhesions, irradiated bowel, short gut, or a history of gastrointestinal cancers.**
- **Cited advantages — vaginal lubrication, achievement of depth, potentially decreased risk of stenosis — have not been extensively studied, so providers must be cautious in offering the procedure as a routine option for primary vaginoplasty.**
- **Disadvantages:** **injury to surrounding abdominal organs during bowel harvest; excessive mucus production causing difficulty with hygiene; diversion colitis; and neovaginal prolapse.**
- **Technique:** an **isolated segment of colon or small intestine**, through a **combined abdominoperineal approach**, the perineal portion as already described; the abdominal portion may be **laparoscopic, robotic or open**. **Adequate mesenteric mobility is essential to pass the bowel into the pelvis without significant tension.**
- **Evolving data with abdominoperineal approaches:** improved visualisation of the proximal canal dissection has reduced **rectal injury to 2.3%, against approximately 7% historically; urethral injury is 1.1%.**
- **Bottom line of the complication review:** **roughly one in eight intestinal vaginoplasties is removed because of postoperative complications, and the large majority require at least one reoperation such as introitoplasty.**
- **Late hazards:** **bothersome mucocoele, and later onset of inflammatory bowel disease.**
- **Dilation remains essential after intestinal vaginoplasty — the vaginal introitus may still close.**

Concept — *The three lining options form an escalating ladder, each entered when the one before runs out of tissue or depth: penile and scrotal skin first, peritoneum when more proximal depth is needed or exposure demands a robot, and bowel only when there is not enough genital skin to line anything. Moving up the ladder buys depth and lubrication but imports the donor organ's own pathology, and never removes the obligation to dilate.*

Common Complications and Incidence

Wound Dehiscence

- **Superficial wound dehiscence is the commonest minor complication after vaginoplasty**, with wound complication rates of **12% to 33%** in systematic review and possibly higher in clinical experience.
- **Most often at the vaginal introitus**, possibly because of **added tension from passing the vaginal dilators**.
- **Managed with local wound care in most cases** — dressings, general hygiene, and **healing by secondary intention**.
- **Granulation tissue within the wound causes bleeding and discharge**, treated with **curettage or cauterisation and topical steroids**.

Rectovaginal Fistula

- **Potentially the most devastating complication**, typically related to **rectal injury during canal dissection**.
- **Incidence 0.8% in primary surgery and 6.3% in revision cases**. Revision must be approached with caution, as the altered anatomy raises the rate of rectal injury.
- **If rectal injury is recognised intraoperatively**: close the rectum in a watertight fashion; cover the site with a bulbospongiosus muscle flap if available; and consider faecal diversion according to the degree of injury. This is why the bulbospongiosus is preserved until canal dissection is complete.
- **No consensus exists on the technique of repair**.
- **The central dilemma: closing a fistula requires rest, but preventing stenosis requires dilation, and dilation traumatises the repair**. Two strategies:
 - **Stop dilation and allow the canal to stenose in anticipation that the fistula will close**, then perform revision surgery for the stenosis. **Drawback: later revision to widen and deepen the canal risks reopening the fistula**.
 - **Defer closure until the canal is sufficiently stable to maintain width and depth**, so the repair can heal without dilation.

Urethrovaginal Fistula

- **Incidence 0.8–3.9%**. Causes: **unrecognised urethral injury during canal dissection, or distal urethral stenosis**. The same balance between closure and dilation applies.
- **Location determines the symptom** — the examinable pair: **fistulas distal to the bladder neck cause vaginal voiding; fistulas at the bladder neck cause incontinence**.
- **Distal fistulas often need no repair**, as vaginal voiding is the only symptom.
- **A proximal fistula is repaired transvaginally**, through mobilisation of the vaginal wall and prostatic urethra.
- **Tissue interposition may be difficult**: the labia majora does not have the blood supply to allow a Martius flap, and the gracilis muscle may obstruct the vaginal introitus.

Urethral Stricture

- **Cited between 1% and 6%**, with the highest study reporting 40%.
- **Presentation: dysuria, weak stream, haematuria, recurrent urinary tract infections, urinary spraying, split stream, or retention**.
- **Workup should include functional aetiologies**, for which pelvic floor physical therapy can be offered.
- **Meatal stenosis may be diagnosed on physical examination alone**; more proximal pathology warrants full workup such as cystoscopy.

- **Dilation may be an appropriate first step for short obstruction, but reconstruction usually involves spatulation and maturation of the urethral meatus.**

Urethral Bulge or Malposition

- **Inadequate debulking of the corpus spongiosum leads to a urethral bulge, which becomes more significant with genital arousal and may obstruct the vaginal canal and cause dyspareunia, pain and dysphoria.**
- **Inadequate mobilisation or dissection of the spongiosum down to the crus of the cavernosa leads to an upward-pointing meatus, causing spraying over the edge of the toilet seat.**
- **Treatment: debulking of the excess tissue and downward fixation of the malpositioned meatus, with closure in multiple layers, to give the stream an appropriate trajectory.**

Vaginal Stenosis

- **Consequence: inability to engage in intercourse.**
- **Causes: poor adherence to dilation, granulation or scar tissue formation, poor graft take, and pelvic floor dysfunction leading to inadequate depth of dilation. Stenosis may occur at the introitus or within the canal.**
- **First step is pelvic floor physical therapy, to maximise canal depth and prevent future pelvic pain from dilation. Steroid cream or injection may be used but has not been well studied.**
- **Surgery only after physical therapy has been completed: excision of the scar with graft use may restore calibre and depth; peritoneal flaps or intestinal augmentation are alternatives.**
- **Emerging experience with allograft in revision surgery: the canal is dissected robotically, lined with allograft, and peritoneal flaps placed at the apex.**
- **A newly dissected revision canal is treated exactly as a primary canal dissection — postoperative packing then dilation.**

Vaginal Prolapse

- **Mechanistically different from prolapse in cisgender women. Not weakness of the pelvic floor, but poor graft take or poor fixation of skin flaps.**
- **Reported incidence 4–10%, although the definition is not uniform and the data derive primarily from perineal approaches.**
- **Incidence may be lower with peritoneal interposition and robotic surgery, because securing the proximal penile and scrotal skin to the peritoneum acts as a colpopexy. Fixation of the canal to surrounding pelvic muscles has been described, for example using a Capio device.**
- **No standard treatment. Early postoperative prolapse may be repacked, but prolapse signifies poor graft take, and vaginal stenosis will often follow even if the canal is repacked.**

Concept — *Nearly every complication traces to one of three failures: a plane violated during canal dissection (rectovaginal and urethrovaginal fistula), tissue inadequately excised or debulked (urethral bulge, erectile engorgement, obstructing bulbospongiosus), or a graft that did not take or was not maintained (dehiscence, stenosis, prolapse). Only the third is shared with the patient, and it is why dilation is counselled as surgical rather than optional.*

Long-Term Outcomes

- **Evidence on long-term function and satisfaction is scant, and until recently there was no transgender-specific patient-reported outcome measure — the principal methodological limitation.**
- **Bottom line of the available series: the large majority report successful sexual arousal, normal genital function, and partner satisfaction with their genitalia.**

- **Orgasmic function:** the large majority are able to orgasm, with a median time to orgasm of 180 days — worth knowing because patients expect it sooner.
- **Neovaginal depth** is reported at 12 to 16 cm (5 to 6.5 inches), against a typical cisgender vaginal depth of 9 to 12 cm (3.5 to 5 inches) — the neovagina is **deeper**, not shallower, than the cisgender norm.
- Some patients report self-lubrication from their peritoneal segments, but surgeons should counsel that this is not the norm.
- **Setting appropriate expectations before surgery is crucial to postoperative satisfaction.**

Concept — *The outcome literature is weakest exactly where counselling matters most — function, lubrication and time to orgasm — so expectation-setting substitutes for evidence. The two figures that most often produce disappointment are the six-month median to orgasm and the assumption of self-lubrication.*

POSTOPERATIVE CARE

Immediate Perioperative Course

- **Counsel before surgery to expect approximately 5 days in hospital.**
- **Maintain subcutaneous heparin, sequential compression devices, broad-spectrum antibiotics, an aggressive bowel regimen and intravenous fluids.**
- **Sitting is permitted, but not prolonged periods with direct pressure over the perineum.**
- **On postoperative day 5, the vaginal packing and the urinary catheter are removed.**
- **Dilation teaching is given by a member of the team** — appropriate technique, duration and signs of complications — together with a **douching protocol after dilations to maintain hygiene.**
- **Urinary spraying is common after the procedure** and generally improves as the **localised tissue oedema resolves.**
- **If urinary retention occurs, replace the catheter. Dilation can be performed with the catheter in place, and the catheter is removed a week later in the outpatient setting.**

Role of Physical Therapy

- **Establish care with a pelvic floor physical therapist before surgery.**
- **Postoperatively, routine visits confirm that dilation is being performed appropriately.**
- **Four to five postoperative visits are adequate for most patients.**

Clinical Follow-up

- Frequent follow-up addresses both **physical recovery from major surgery** and **a host of new sentiments related to the genitalia**, which generate logistical questions and a need for emotional support.
- **At each visit: physical examination of the external genitalia for healing, and speculum examination of the canal for depth and excessive granulation tissue.**
- **Significant granulation tissue is removed with a curette or sharply excised;** if persistent, **low-dose steroid ointment placed on the tip of the dilator** delivers treatment to the proximal canal.
- **For foul odour or concern for bacterial infection, an antibiotic gel is used in the same way.**
- **Inadequate preoperative hair removal manifests here** — regrowth of hair in the canal requiring **intermittent removal to maintain hygiene.**
- **Minor wound dehiscence, especially at the six o'clock position of the canal, is common and is successfully managed with local wound care.**
- **Example protocol: weekly for the first 3 weeks, then 6 weeks, 3 months, 6 months, 12 months, and annually thereafter.**

Importance of Dilation

- **The neovaginal canal may take months to years to heal fully**, and even then **may be physiologically perceived by the body as a pelvic defect** — hence a **lifelong risk of canal stenosis** if **adequate dilation is not maintained**.
- **An aggressive early dilation protocol is crucial to maintaining depth and calibre**. Example: **four times daily for 15 to 20 minutes for the first 6 weeks; three times daily for 15 to 20 minutes until 3 months; twice daily for 15 to 20 minutes from 3 months to 1 year; then once daily**.
- **After 1 year the patient may dilate less frequently if depth and calibre are not being lost**.
- **At any sign of canal stenosis, increase the frequency of dilation and contact the surgeon**, as **additional pelvic floor physical therapy may be warranted to regain depth and calibre**.

Concept — *The operation creates a space the body treats as a wound to be closed, so the canal is maintained rather than healed — a lifelong mechanical obligation that tapers in frequency but never ends. This is why non-adherence risk is assessed by mental health services before surgery, and why postoperative depression counts as a surgical complication here.*